Bangla → IPA: Full Reference & Conversion Guide

বাংলা · INTERNATIONAL PHONETIC ALPHABET

VOWELS | VOWEL SIGNS | STOPS | NASALS & OTHERS | SPECIAL SIGNS | STEP-BY-STEP

1 · INDEPENDENT VOWELS · স্বরবর্ণ

SCRIPT	NAME	IPA	HEIGHT · BACKNESS	EXAMPLE WORD	IPA TRANSCRIPTION
অ	ô / shôr ô	/ɔ/	Open-mid · back unrounded	অনেক	/ɔnek/
আ	aa / shôr aa	/a/	Open · central unrounded	আম	/am/
ই	hrôshô i	/i/	Close · front unrounded	ইট	/iʈ/
ঈ	dîrgho i	/i/	Close · front (same as ই in modern Bangla)	ঈদ	/id/
উ	hrôshô u	/u/	Close · back rounded	উট	/uʈ/
ঊ	dîrgho u	/u/	Close · back rounded (same as উ)	ঊন	/un/
ঋ	ri / shôr ri	/ri/	Syllabic rhotic sequence	ঋতু	/ritu/
এ	ê / shôr ê	/e/ ~ /ɛ/	Close-mid to open-mid · front	এক	/ek/
ঐ	oi	/oj/	Diphthong: /o/ + /j/ glide	ঐক্য	/okkʰo/
ও	o / shôr o	/o/	Close-mid · back rounded	ওষুধ	/oʃud̪ʰ/
ঔ	ou	/ow/	Diphthong: /o/ + /w/ glide	ঔষধ	/owʃod̪ʰ/

2 · VOWEL SIGNS (MATRAS) · কারচিহ্ন

SIGN	NAME	IPA	POSITION	EXAMPLE	FULL IPA
ক + া = কা	aa-kaar	/a/	Right of consonant	কাজ	/kadʒ/
কি	hrôshô i-kaar	/i/	Left of consonant (written before)	কিছু	/kitʃʰu/
কী	dîrgho i-kaar	/i/	Right of consonant	কীভাবে	/kibʰabe/
কু	hrôshô u-kaar	/u/	Below consonant	কুকুর	/kukur/
কূ	dîrgho u-kaar	/u/	Below consonant	কূল	/kul/
কৃ	ri-kaar	/ri/	Below consonant	কৃষক	/kriʃɔk/
কে	ê-kaar	/e/	Left of consonant	কেন	/ken/
কৈ	oi-kaar	/oj/	Left of consonant	কৈশোর	/kojʃor/
কো	o-kaar	/o/	Both sides of consonant	কোথায়	/kotʰaj/
কৌ	ou-kaar	/ow/	Both sides of consonant	কৌশল	/kowʃɔl/

3 · STOPS / PLOSIVES — স্পর্শবর্ণ (4-WAY CONTRAST)

SCRIPT	NAME	IPA	PLACE	VOICING / ASPIRATION	EXAMPLE	FULL IPA
প	pô	/p/	Bilabial	Voiceless · unaspirated	পাখি	/pakʰi/
ফ	phô	/pʰ/ ~ /f/	Bilabial	Voiceless · aspirated	ফুল	/pʰul/

ব	bô	/b/	Bilabial	Voiced · unaspirated	বাড়ি	/baɽi/
ভ	bhô	/bʰ/ ~ /v/	Bilabial	Voiced · aspirated	ভালো	/bʰalo/
ত	tô	/t̪/	Dental	Voiceless · unaspirated	তাল	/t̪al/
থ	thô	/t̪ʰ/	Dental	Voiceless · aspirated	থালা	/t̪ʰala/
দ	dô	/d̪/	Dental	Voiced · unaspirated	দাদা	/d̪ad̪a/
ধ	dhô	/d̪ʰ/	Dental	Voiced · aspirated	ধান	/d̪ʰan/
ট	ṭô	/ʈ/	Retroflex	Voiceless · unaspirated	টাকা	/ʈaka/
ঠ	ṭhô	/ʈʰ/	Retroflex	Voiceless · aspirated	ঠান্ডা	/ʈʰanda/
ড	ḍô	/ɖ/	Retroflex	Voiced · unaspirated	ডাল	/ɖal/
ঢ	ḍhô	/ɖʰ/	Retroflex	Voiced · aspirated	ঢাকা	/ɖʰaka/
চ	cô	/tʃ/	Palatal	Voiceless · unaspirated	চাল	/tʃal/
ছ	chô	/tʃʰ/	Palatal	Voiceless · aspirated	ছাতা	/tʃʰata/
জ	jô	/dʒ/	Palatal	Voiced · unaspirated	জল	/dʒɔl/
ঝ	jhô	/dʒʰ/	Palatal	Voiced · aspirated	ঝড়	/dʒʰɔɽ/
ক	kô	/k/	Velar	Voiceless · unaspirated	কলম	/kɔlɔm/
খ	khô	/kʰ/	Velar	Voiceless · aspirated	খাবার	/kʰabar/
গ	gô	/g/	Velar	Voiced · unaspirated	গাছ	/gatʃ/
ঘ	ghô	/gʰ/	Velar	Voiced · aspirated	ঘর	/gʰɔr/

4 · NASALS · FRICATIVES · LIQUIDS · APPROXIMANTS

NASALS

SCRIPT	IPA	PLACE	EXAMPLE	FULL IPA
ঙ	/ŋ/	Velar	বাংলা	/baŋla/
ঞ	/n/	Palatal (rare alone)	জ্ঞান	/dʒnan/
ণ	/n/	Retroflex (→ /n/ colloquially)	কণা	/kɔna/
ন	/n/	Dental	নাম	/nam/
ম	/m/	Bilabial	মাছ	/matʃ/

FRICATIVES

SCRIPT	IPA	PLACE	EXAMPLE	FULL IPA
শ	/ʃ/	Palato-alveolar	শাল	/ʃal/
ষ	/ʃ/	Retroflex (→ /ʃ/ colloquially)	ষাট	/ʃaʈ/
স	/s/	Dental	সাপ	/sap/
হ	/h/	Glottal	হাত	/hat/

LIQUIDS & APPROXIMANTS

SCRIPT	IPA	TYPE	EXAMPLE	FULL IPA
র	/r/	Alveolar trill/tap	রাত	/rat/
ল	/l/	Lateral approximant	লাল	/lal/
য	/dʒ~j/	Initial /dʒ/; medial /j/	যাওয়া	/dʒaoa/
ব	/b~w/	Initial /b/; medial /w/	স্বাধীন	/ʃwad̪ʰin/

5 · SPECIAL SIGNS & MODIFIED CONSONANTS

SIGN	NAME	IPA	FUNCTION	EXAMPLE	FULL IPA
ড়	ṛô (dot-ra)	/ɽ/	Retroflex flap (not a stop)	গড়া	/gɔɽa/
ঢ়	ṛhô (dot-ḍha)	/ɽʰ/	Aspirated retroflex flap	rare	/ɽʰ/
য়	ântastho yô	/j/	Palatal glide (not affricate)	নয়	/nɔj/
◌ং	anusvara	/ŋ/	Velar nasal coda; never onset	বাংলা	/baŋla/
◌ঃ	visarga	/h/	Voiceless aspirate after vowel	দুঃখ	/d̪ukʰ/
◌ঁ	chandrabindu	/◌̃/	Nasalises the preceding vowel	চাঁদ	/tʃãd̪/
◌্	hasanta / virāma	(silence)	Kills the inherent /ɔ/; creates conjunct	ক্ + ষ = ক্ষ	/kʃ/

6 · HOW TO CONVERT A BANGLA WORD TO IPA — STEP BY STEP

Core rule to memorise: Every consonant carries an inherent vowel /ɔ/ unless it is (a) followed by a matra/vowel sign, or (b) marked with hasanta ◌্, or (c) at word-final position where /ɔ/ is often dropped in standard Dhaka speech.

1 Segment the word into graphemes

Break the word into individual units: independent vowels, consonants, matras, and diacritics. Watch for conjuncts (যুক্তবর্ণ) — two consonants joined by hasanta ◌্.

বাং (ব + ◌া + ◌ং) + লা (ল + ◌া) + দে (দ + ে◌) + শ (শ (bare))

WORD: বাংলাদেশ

2 Map each grapheme to its IPA symbol

Look up each consonant and vowel/matra in the tables above.

GRAPHEME	TYPE	IPA	NOTE
ব	Consonant	/b/	Bilabial voiced stop
◌া	Matra (aa-kaar)	/a/	Replaces inherent /ɔ/
◌ং	Anusvara	/ŋ/	Velar nasal coda
ল	Consonant	/l/	Lateral approximant
◌া	Matra (aa-kaar)	/a/	Replaces inherent /ɔ/
দ	Consonant	/d̪/	Dental voiced stop
ে◌	Matra (ê-kaar)	/e/	Replaces inherent /ɔ/
শ	Consonant (word-final)	/ʃ/	Final /ɔ/ dropped

3 Apply the inherent vowel rule

Each bare consonant (no matra, no hasanta) gets an inherent /ɔ/ — *except* when it is word-final (inherent vowel is dropped in standard Dhaka Bangla).

Rule: consonant + (no matra, not word-final) → append /ɔ/
Rule: consonant + (no matra, word-final) → no vowel appended
Rule: consonant + matra → matra's vowel replaces /ɔ/
Rule: consonant + hasanta (◌্) → no vowel at all (creates conjunct)

ব /b/ +ো → /a/ → /ba/ ✓ matra + ং /ŋ/ coda + ল /l/ +ো → /a/ → /la/ ✓ matra + দ /d̪/ +ে → /e/ → /d̪e/ ✓ matra + শ /ʃ/ final→no ɔ

4

Check vowel harmony

Look for /ɔ/ in non-final syllables. If the following syllable contains a high or mid vowel (/i/, /u/, /e/, /o/), raise /ɔ/ → /o/. If the following syllable has /a/, /ɔ/ stays.

> In **বাংলাদেশ**, all vowels are provided by matras (আ, এ) or are coda consonants (ং), so no inherent /ɔ/ exists to raise. Harmony is already satisfied.

Harmony example: **বলো** — root /bɔl/ + suffix /o/ → /bolo/ (raising applies).

5

Handle conjuncts, chandrabindu, anusvara

Conjuncts (hasanta joins two consonants): drop the inherent vowel between them and merge their IPA directly.
Chandrabindu ঁ: add tilde to the vowel → /ã/, /ẽ/, etc.

Anusvara ং: → /ŋ/ as a syllable coda.

EXAMPLE	BREAKDOWN	IPA
ক্ষ	ক্ + ষ (hasanta joins)	/kʃ/
চাঁদ	চ + া + ঁ + দ	/tʃãd̪/
বাংলা	ব + া + ং + ল + া	/baŋla/
স্কুল	স + ্ + ক + ু + ল	/skul/

6

Assemble the final transcription

Concatenate all segments in order and enclose in slashes for phonemic notation.

/b/ + /a/ + /ŋ/ + /l/ + /a/ + /d̪/ + /e/ + /ʃ/

RESULT: **বাংলাদেশ** → **/baŋlad̪eʃ/**

SECOND EXAMPLE

মা /ma/ ম+ো + নু /nu/ ন+ু + ষ /ʃ/ final

RESULT: **মানুষ** → **/manuʃ/**

THIRD EXAMPLE — INHERENT /ɔ/ APPEARS

ক /k/ bare → +ɔ + ল /l/ bare → +ɔ + ম /m/ final

RESULT: **কলম** → **/kɔlɔm/**

FOURTH EXAMPLE — CONJUNCT CONSONANT (যুক্তবর্ণ)

Step A · Segment — identify graphemes left to right

ব /b/ consonant + ি /i/ matra + জ /dʒ/ 1st of conjunct + ্ (ø) hasanta + ঞ /n/ 2nd of conjunct + া /a/ matra + ন /n/ final

Hasanta rule: জ + ্ (hasanta) + ঞ → the hasanta kills the inherent /ɔ/ after জ and merges the two consonants into the cluster /dʒn/. No vowel appears between them.

Step B · Map to IPA

GRAPHEME	TYPE	IPA	RULE APPLIED
ব	Consonant	/b/	Normal bilabial stop
ি	Matra (hrôshô i-kaar)	/i/	Replaces inherent /ɔ/ of ব
জ	Consonant — 1st of conjunct	/dʒ/	Hasanta follows → no inherent vowel
্	Hasanta (virama)	ø	Suppresses vowel; fuses জ + ঞ into cluster
ঞ	Consonant — 2nd of conjunct	/n/	Follows hasanta; gets next vowel (া)
া	Matra (aa-kaar)	/a/	Attaches to ঞ, replaces its inherent /ɔ/
ন	Consonant — word-final	/n/	Final position → inherent /ɔ/ dropped

Step C · Assemble

/b/ + /i/ + /dʒ/ ‿ /n/ + /a/ + /n/

RESULT: বিজ্ঞান → /bidʒnan/ "science"

More common conjuncts

CONJUNCT	COMPONENTS	IPA CLUSTER	EXAMPLE WORD	FULL IPA
ক্ষ	ক + ্ + ষ	/kʃ/	ক্ষমা	/kʃɔma/
ত্র	ত + ্ + র	/t̪r/	পিতা	/t̪ra/
প্র	প + ্ + র	/pr/	প্রতি	/prɔti/
স্ক	স + ্ + ক	/sk/	স্কুল	/skul/
ন্ত	ন + ্ + ত	/nt̪/	অন্ত	/ɔnt̪/
ষ্ট	ষ + ্ + ট	/ʃʈ/	কষ্ট	/kɔʃʈ/

7 Chandrabindu nasalization — ঁ

The chandrabindu ঁ sits above a consonant+vowel unit and nasalizes the vowel of that syllable. It does *not* add a consonant — it only changes the vowel from oral to nasal. In IPA a tilde ◌̃ is placed on the affected vowel.

Step A · Segment

চ /tʃ/ consonant + া /a/ matra + ঁ nasalize chandrabindu + দ /d̪/ final

Chandrabindu rule: ঁ retroactively nasalizes the vowel in the same syllable. চ + া = /tʃa/ → with ঁ → /tʃã/. The chandrabindu itself contributes no IPA segment — only the tilde on the vowel.

Step B · Map to IPA

GRAPHEME	TYPE	IPA	RULE APPLIED
চ	Consonant	/tʃ/	Palatal voiceless affricate

◌া	Matra (aa-kaar)	/a/ → /ã/	Replaces inherent /ɔ/; then nasalized by ◌ঁ
◌ঁ	Chandrabindu	◌̃	Nasalizes the vowel of its syllable — no IPA segment of its own
দ	Consonant — word-final	/d̪/	Dental voiced stop; final → inherent /ɔ/ dropped

Step C · Assemble

/tʃ/ + /ã/ nasalized + /d̪/

RESULT: চাঁদ → /tʃãd̪/ "moon"

Oral vs. nasal contrast — same base vowel, different nasality

WORD	IPA	MEANING	NASALIZED VOWEL
চাদ	/tʃad̪/	sheet (cloth)	oral /a/
চাঁদ	/tʃãd̪/	moon	nasal /ã/ — meaning changes!

More chandrabindu examples across different vowels

WORD	BREAKDOWN	NASAL VOWEL	FULL IPA	MEANING
হাঁস	হ + ◌া + ◌ঁ + স	/ã/	/hãs/	duck
বাঁশ	ব + ◌া + ◌ঁ + শ	/ã/	/bãʃ/	bamboo
কাঁদা	ক + ◌া + ◌ঁ + দ + ◌া	/ã/	/kãda/	to cry
মেঁও	ম + ◌ে + ◌ঁ + ও	/ẽ/	/mẽo/	meow
গোঁফ	গ + ◌ো + ◌ঁ + ফ	/õ/	/gõpʰ/	moustache
কুঁড়ি	ক + ◌ু + ◌ঁ + ড় + ি	/ũ/	/kũɽi/	bud (flower)

Key distinctions:

· Chandrabindu ◌ঁ nasalizes a *vowel* — contrast with anusvara ◌ং which adds a velar nasal *consonant* /ŋ/ as a coda.

· In fast/colloquial speech, chandrabindu nasalization may weaken or merge with a following nasal consonant.

· Nasality is phonemically contrastive in Bangla: oral vs. nasal vowel pairs can distinguish meaning (চাদ vs. চাঁদ above).

8

Anusvara — ◌ং → /ŋ/

The anusvara ◌ং always contributes a velar nasal consonant /ŋ/ as a *syllable coda* — it closes the syllable before it and never begins one. Unlike chandrabindu ◌ঁ, it adds a real consonant segment to the IPA, not just a feature on the vowel.

Step A · Segment

র /r/ consonant + ◌ং /ŋ/ anusvara coda

Anusvara rule: ◌ং always functions as a syllable *coda* — it closes the syllable of the consonant before it. Since ◌ং is not a vowel or matra, the preceding consonant র still carries its inherent /ɔ/. So: র → /rɔ/ (inherent vowel) + ◌ং → /ŋ/ (coda) = /rɔŋ/.

Step B · Map to IPA

GRAPHEME	TYPE	IPA	RULE APPLIED
র	Consonant	/r/ + inherent /ɔ/	No matra; ◌ং is a coda, not a vowel → inherent /ɔ/ applies
◌ং	Anusvara	/ŋ/	Velar nasal consonant — coda of syllable /rɔ/; never an onset

Step C · Assemble

/r/ + /ɔ/ inherent + /ŋ/ coda

RESULT: রং → /rɔŋ/ "colour"

Second example — anusvara mid-word: বাংলা

ব /b/ consonant + া /a/ matra + ং /ŋ/ coda + ল /l/ consonant + া /a/ matra

Syllable structure: /baŋ/ · /la/ — ং closes the first syllable, ল opens the second.

RESULT: বাংলা → /baŋla/ "Bangla"

Critical contrast: anusvara ং vs. chandrabindu ঁ

SIGN	IPA CONTRIBUTION	AFFECTS	EXAMPLE	IPA	MEANING
ং anusvara	/ŋ/ — adds a consonant	Closes the syllable as a velar nasal coda	বাং	/baŋ/	Bangla (partial)
ঁ chandrabindu	◌̃ — nasalizes the vowel	Adds nasality feature to the vowel; no extra consonant	বাঁ	/bã/	left (direction)

More anusvara examples

WORD	BREAKDOWN	SYLLABLE STRUCTURE	FULL IPA	MEANING
আংটি	আ + ং + ট + ি	/aŋ/ · /ʈi/	/aŋʈi/	ring
ঢং	ঢ + ং	/ɖʰɔŋ/	/ɖʰɔŋ/	style / manner
সংসার	স + ং + স + া + র	/sɔŋ/ · /sar/	/sɔŋsar/	family / world
অংক	অ + ং + ক	/ɔŋ/ · /k/	/ɔŋk/	number / digit
বংশ	ব + ং + শ	/bɔŋ/ · /ʃ/	/bɔŋʃ/	lineage / clan
মঙ্গল	ম + ং + গ + ল	/mɔŋ/ · /gɔl/	/mɔŋgɔl/	auspicious / Mars

Three rules for anusvara:

· /ŋ/ is always a *coda* — it can never open a syllable in Bangla.

· It does *not* replace the inherent vowel of the preceding consonant; that vowel still applies normally.

· When ং appears before a consonant, the coda /ŋ/ closes the previous syllable and the next consonant opens a new one (বাং·লা, সং·সার).

9

Visarga — ঃ → /h/

The visarga ঃ always follows a vowel and contributes a voiceless glottal fricative /h/. It is a coda element like anusvara, but phonetically it is a consonant of breath — not a nasal. When it precedes an aspirated stop (খ, ঘ, ছ, ঝ, ঠ, ঢ, থ, ধ, ফ, ভ), its /h/ merges into that stop's own aspiration.

Step A · Segment — দুঃখ

দ /d̪/ consonant + ু /u/ matra + ঃ /h/ visarga + খ /kʰ/ final

Visarga rule: ঃ → /h/ as a post-vowel coda. When ঃ immediately precedes an *aspirated stop* (here খ = /kʰ/), the visarga's /h/ and the stop's own aspiration ʰ merge into a single aspirate: /h/ + /kʰ/ → /kʰ/. The aspiration is not doubled — it is

absorbed.

Step B · Map to IPA

GRAPHEME	TYPE	IPA	RULE APPLIED
দ	Consonant	/d̪/	Dental voiced stop
ু	Matra (hrôshô u-kaar)	/u/	Replaces inherent /ɔ/ of দ
ঃ	Visarga	/h/	Post-vowel /h/ coda; merges with following খ's aspiration → not heard separately
খ	Consonant — word-final	/kʰ/	Velar voiceless aspirated; final → no inherent /ɔ/

Step C · Assemble

/d̪/ + /u/ + /h/ absorbed → ◠ /kʰ/

RESULT: দুঃখ → /d̪ukʰ/ "sorrow / pain"

Second example — visarga before unaspirated consonant: অতঃপর

অ /ɔ/ vowel + ত /t̪/ +inherent ɔ + ঃ /h/ stays /h/ + প /p/ +inherent ɔ + র /r/ final

Here প is *unaspirated*, so ঃ's /h/ has nothing to merge into — it is pronounced as a distinct breathy /h/ between the syllables: /ɔt̪ɔ/ · /hpɔr/.

RESULT: অতঃপর → /ɔt̪ɔhpɔr/ "thereafter"

Merger vs. no merger — the key split

CONTEXT	BEHAVIOUR	EXAMPLE	IPA
ঃ before aspirated stop (খ ঘ ছ ঝ ঠ ঢ থ ধ ফ ভ)	/h/ absorbed into stop's aspiration	দুঃখ	/d̪ukʰ/
ঃ before unaspirated consonant	/h/ pronounced as distinct segment	অতঃপর	/ɔt̪ɔhpɔr/
ঃ word-finally (after vowel)	/h/ audible as final breathy release	স্বতঃ	/ʃʷɔtɔh/

More visarga examples

WORD	BREAKDOWN	FULL IPA	MEANING	NOTE
পুনঃপুনঃ	পুন + ঃ + পুন + ঃ	/punɔhpunɔh/	again and again	/h/ before প stays
ক্রমশঃ	ক্রমশ + ঃ	/krɔmɔʃɔh/	gradually	word-final /h/
প্রায়শঃ	প্রায়শ + ঃ	/praɔʃɔh/	frequently	word-final /h/
নিঃশব্দ	নি + ঃ + শব্দ	/niʃɔbd̪/	silent / soundless	/h/ + /ʃ/ → /ʃ/ (partial merger)

Three visarga rules:

· ঃ always follows a vowel and always contributes /h/ — it never begins a syllable.

· Before an *aspirated* stop its /h/ merges into the stop's own aspiration (দুঃখ → /d̪ukʰ/, not /d̪uhkʰ/).

· In all other positions — before an unaspirated consonant, or word-finally — the /h/ is a distinct audible segment.

10

Hasanta / Virāma — ্ (vowel killer)

The hasanta ্ sits below a consonant and *kills its inherent /ɔ/*, leaving that consonant bare with no vowel. The bare consonant then fuses with the next consonant to form a conjunct cluster. Without hasanta, each consonant would get its own inherent /ɔ/, producing a completely different word.

Step A · Segment — শক্তি

শ	+	ক	+	্	+	ত	+	ি
/ʃ/ consonant		/k/ 1st of conjunct		kills /ɔ/ hasanta		/t̪/ 2nd of conjunct		/i/ matra

Hasanta rule: ক would normally carry inherent /ɔ/, giving /kɔ/. The hasanta ্ underneath it strips that /ɔ/ away, leaving bare /k/. That bare /k/ then attaches directly to ত, forming the cluster /kt̪/. The matra ি then supplies the vowel for ত: /t̪i/.

Step B · Map to IPA — with vs. without hasanta

GRAPHEME	WITHOUT HASANTA	WITH HASANTA ্	WHY
শ	/ʃɔ/	/ʃɔ/	No matra; inherent /ɔ/ applies in both cases
ক + ্	/kɔ/ (no hasanta)	/k/ ← bare	Hasanta kills the inherent /ɔ/ of ক
ত + ি	/t̪ɔ/ (no matra)	/t̪i/	Matra ি replaces inherent /ɔ/ of ত

Step C · Assemble

/ʃ/	+	/ɔ/ inherent	+	/k/ bare (্)	‿	/t̪/	+	/i/ matra

RESULT: **শক্তি** → **/ʃɔkt̪i/** "power / energy"

Critical contrast — hasanta changes meaning entirely

WRITTEN FORM	BREAKDOWN	IPA	MEANING
শক্তি	শ + ক + ্ + ত + ি	/ʃɔkt̪i/	power / energy ✓ (with hasanta)
শকতি	শ + ক + ত + ি (no ্)	/ʃɔkɔt̪i/	not a real word ✗ (without hasanta, each consonant gets /ɔ/)
কষ্ট	ক + ষ + ্ + ট	/kɔʃʈ/	suffering ✓ (with hasanta)
কষট	ক + ষ + ট (no ্)	/kɔʃɔʈ/	not a real word ✗ (3 separate syllables)

Word-final hasanta — explicit /ɔ/ drop

In modern colloquial Bangla the final inherent /ɔ/ is already dropped without any written marker. In classical or Sanskrit-origin words, a word-final hasanta is sometimes written explicitly to show the consonant carries no vowel at all.

WORD	BREAKDOWN	FULL IPA	MEANING	HASANTA POSITION
মহৎ	ম + হ + ৎ	/mɔhɔt/	noble / great	ৎ (khanda ta) = final /t/ with no vowel
রাত্রি	র + া + ত + ্ + র + ি	/raṯri/	night	mid-word ্ kills ত's /ɔ/; র gets matra ি
শক্ত	শ + ক + ্ + ত	/ʃɔkt/	hard / firm	mid-word ্; final ত drops /ɔ/ (colloquial)
মুক্ত	ম + ু + ক + ্ + ত	/mukt/	free / liberated	্ after ক; ু matra on ম; final ত bare
সত্য	স + ত + ্ + য	/ʃɔt̪ʈɔ/	truth	্ kills ত's /ɔ/; য → /ʈɔ/ (gemination in some dialects)

Three hasanta rules:

· ্ strips the inherent /ɔ/ from the consonant it sits under — that consonant becomes bare.

· The bare consonant fuses with the next consonant, forming a cluster written as a single conjunct glyph.

· A matra on the *second* consonant of the conjunct gives that consonant its vowel normally; the *first* consonant has none.

11

The Inherent Vowel Rule — /ɔ/ by default

Every Bangla consonant that is written bare — with no matra, no hasanta, and no following vowel letter — carries the inherent vowel **/ɔ/** automatically. This is the single most important rule in Bangla-to-IPA conversion. The only exception is the *word-final* position, where standard Dhaka Bangla drops the inherent /ɔ/ in normal speech.

Step A · Segment — গরম (three bare consonants)

গ	+	র	+	ম
/g/		/r/		/m/
bare · non-final		bare · non-final		bare · **final**

Inherent vowel rule:
গ — bare, non-final → append **/ɔ/** → **/gɔ/**
র — bare, non-final → append **/ɔ/** → **/rɔ/**
ম — bare, **word-final** → **/ɔ/** dropped → **/m/**

Step B · Map to IPA — position determines whether /ɔ/ appears

#	GRAPHEME	TYPE	POSITION	IPA	REASON
1	গ	Consonant — bare	Non-final	/gɔ/	No matra, no hasanta → inherent /ɔ/ added
2	র	Consonant — bare	Non-final	/rɔ/	No matra, no hasanta → inherent /ɔ/ added
3	ম	Consonant — bare	Word-final	/m/	Final position → inherent /ɔ/ dropped in standard speech

Step C · Assemble

/gɔ/	+	/rɔ/	+	/m/
+inherent ɔ		+inherent ɔ		final — no ɔ

RESULT: **গরম** → **/gɔrɔm/** "hot / warm"

The 4-way decision for every consonant you encounter

CONSONANT HAS…	VOWEL RESULT	EXAMPLE	IPA
Nothing (bare, non-final)	**/ɔ/** appended	গ in গরম	/gɔ/
Nothing (bare, word-final)	**/ɔ/** dropped	ম in গরম	/m/
A matra (vowel sign)	Matra's vowel replaces /ɔ/	কা in কাজ	/ka/
Hasanta ্ beneath it	/ɔ/ killed entirely — bare consonant	ক্ in শক্তি	/k/

Contrast — same root consonants, different vowel treatment

WORD	VOWEL TREATMENT	IPA	MEANING
কর	ক bare non-final → /kɔ/ · র bare final → /r/	/kɔr/	do! (imperative)
করো	ক bare → /kɔ/ · র + ো matra → /ro/	/koro/	please do (polite)
করা	ক bare → /kɔ/ · র + া matra → /ra/	/kɔra/	to do (infinitive)
করি	ক bare → /kɔ/ · র + ি matra → /ri/	/kɔri/	I do (present)

More examples — counting inherent /ɔ/ positions

WORD	BARE CONSONANTS	INHERENT /ɔ/ COUNT	FULL IPA	MEANING
জল	জ (non-final) · ল (final)	1 × /ɔ/ on জ	/dʒɔl/	water
বনাম	ব (non-final) · ন+া matra · ম (final)	1 × /ɔ/ on ব	/bɔnam/	versus
কলম	ক (non-final) · ল (non-final) · ম (final)	2 × /ɔ/ on ক and ল	/kɔlɔm/	pen
সমতল	স · ম · ত (all non-final) · ল (final)	3 × /ɔ/ on স, ম, ত	/ʃɔmɔt̪ɔl/	flat / plain

মনমতন	ম · ন (non-final) + ম · ত · ন (final)	4 × /ɔ/	/mɔnmɔt̪ɔn/	as one wishes

Three inherent vowel rules:

· A bare consonant in *non-final* position always carries /ɔ/ — this is never written, only inferred.

· A bare consonant in *word-final* position drops the /ɔ/ in standard Dhaka Bangla (but may retain it in formal, poetic, or regional speech).

· A matra or hasanta overrides the inherent vowel completely — these two markers are the only way to suppress or replace /ɔ/.

12

Vowel Harmony — /ɔ/ → /o/ raising

After the inherent /ɔ/ is assigned (Step 11), check whether it needs to be raised. If a syllable carries /ɔ/ and the *following syllable* contains a high or mid vowel (/i/, /u/, /e/, /o/), the /ɔ/ raises to /o/. If the following syllable contains the low vowel /a/ or another /ɔ/, raising is blocked and /ɔ/ stays.

Step A · Segment — করি

ক	+	র	+	ি
/k/ + /ɔ/? bare · non-final		/r/ consonant		/i/ trigger vowel

Harmony check: ক carries inherent /ɔ/ → look at the next syllable. র + ি = /ri/, which contains the high vowel /i/. That triggers raising: /ɔ/ → /o/. So ক becomes /ko/, not /kɔ/.

Step B · Map to IPA — with harmony check

GRAPHEME	INHERENT VOWEL	FOLLOWING SYLLABLE	HARMONY?	FINAL IPA
ক	/ɔ/ (bare, non-final)	/ri/ — contains /i/	✓ Raises	/ko/ ← raised from /kɔ/
র + ি	Matra ি → /i/	word-final	—	/ri/

Step C · Assemble

/ko/ raised from /kɔ/	+	/ri/

RESULT: **করি** → **/kori/** "I do"

Minimal pair — same root ক + র, different suffix vowel, different harmony outcome

WORD	SUFFIX VOWEL	HARMONY?	ROOT /ɔ/ BECOMES	FULL IPA	MEANING
করি	/i/ — high	✓ Raises	/o/	/kori/	I do
করু	/u/ — high	✓ Raises	/o/	/koru/	do! (intimate)
করে	/e/ — mid	✓ Raises	/o/	/kore/	does / doing
করো	/o/ — mid	✓ Raises	/o/	/koro/	please do (polite)
করা	/a/ — low	✗ Blocked	/ɔ/	/kɔra/	to do (infinitive)
কর	none (final bare)	✗ Blocked	/ɔ/	/kɔr/	do! (abrupt)

Multi-syllable words — harmony scans one syllable ahead

WORD	BREAKDOWN	WHICH /ɔ/ RAISES?	FULL IPA	MEANING
বলতে	বল + তে	/bɔl/ → /bol/ (তে has /e/); ত+ে → /te/	/bolte/	to say (verbal noun)
বলতি	বল + তি	/bɔl/ → /bol/ (তি has /i/)	/bolti/	spoken language
গরমে	গরম + এ	গ: next syl /rɔ/ has /ɔ/ → no raise; র: next syl /me/ has /e/ → raises	/gɔrome/	in the heat

কবিতা	কবিতা	ক: next syl /bi/ has /i/ → raises → /ko/; rest have matras	/kobita/	poem

Vowel harmony in four rules:

· **Trigger vowels** (/i/, /u/, /e/, /o/) in the following syllable → raise /ɔ/ → /o/ .

· **Blocking vowels** (/a/, /ɔ/) in the following syllable → /ɔ/ stays.

· Harmony is *progressive*: scan left-to-right, one syllable at a time — each /ɔ/ checks only the syllable immediately after it.

· Harmony applies only to the *inherent* /ɔ/; vowels supplied by matras are never raised by this rule.

13

Word-Final Vowel Drop — bare final consonant loses /ɔ/

A bare consonant in *word-final* position would theoretically carry inherent /ɔ/, but standard Dhaka Bangla drops it completely in normal speech. This is one of the most consistent phonological rules in modern Bangla and affects nearly every word that ends in a bare consonant. Adding even a single suffix shifts that consonant away from final position and the /ɔ/ rule applies normally again.

Step A · Segment — ঘর

ঘ /gʰ/ bare · non-final + র /r/ only bare · FINAL

Word-final drop rule: ঘ is non-final (র follows) → gets inherent /ɔ/ → /gʰɔ/ . র is word-final → the inherent /ɔ/ is **dropped** → bare /r/ , not /rɔ/ . Without this rule the word would wrongly be /gʰɔrɔ/ .

Step B · Map to IPA

GRAPHEME	POSITION	INHERENT /ɔ/?	FINAL IPA	WHY
ঘ	Non-final	✓ Kept → /ɔ/	/gʰɔ/	র follows, so ঘ is not final → /ɔ/ applies
র	Word-final	✗ Dropped	/r/	Last consonant in word → inherent /ɔ/ suppressed

Step C · Assemble

/gʰɔ/ +inherent ɔ + /r/ /ɔ/ dropped

RESULT: **ঘর** → **/gʰɔr/** not ✗ /gʰɔrɔ/ · "house / room"

Adding a suffix un-finalizes the consonant — /ɔ/ rule reactivates

FORM	BREAKDOWN	র POSITION	র VOWEL	FULL IPA	MEANING
ঘর	ঘ + র (bare)	Final	dropped	/gʰɔr/	house (base form)
ঘরে	ঘ + র + ে matra	Non-final	matra ে → /e/	/gʰɔre/	in the house (locative)
ঘরের	ঘ + র + ে + র	Non-final (mid)	matra ে → /e/	/gʰɔrer/	of the house (genitive)
ঘরটা	ঘ + র + ট + া	Non-final	inherent /ɔ/	/gʰɔrʈa/	the house (definite)
ঘরগুলো	ঘ + র + গুলো	Non-final	inherent /ɔ/	/gʰɔrgulo/	the houses (plural)

Standard Dhaka vs. formal / poetic speech

WORD	STANDARD / COLLOQUIAL	FORMAL / POETIC	MEANING
ঘর	/gʰɔr/ final /ɔ/ dropped	/gʰɔrɔ/ (retained)	house
মন	/mɔn/	/mɔnɔ/	mind / heart
বন	/bɔn/	/bɔnɔ/	forest
পথ	/pɔt̪ʰ/	/pɔt̪ʰɔ/	path / road

More word-final drop examples across consonant types

WORD	FINAL CONSONANT	WRONG (WITHOUT DROP)	CORRECT IPA	MEANING
গাছ	চ → /tʃ/	✗ /gatʃɔ/	/gatʃ/	tree
হাত	ত → /t̪/	✗ /hat̪ɔ/	/hat̪/	hand
নাম	ম → /m/	✗ /namɔ/	/nam/	name
চোখ	খ → /kʰ/	✗ /tʃokʰɔ/	/tʃokʰ/	eye
রাত	ত → /t̪/	✗ /rat̪ɔ/	/rat̪/	night
ভাত	ত → /t̪/	✗ /bʰat̪ɔ/	/bʰat̪/	rice (cooked)

Three word-final drop rules:

· A bare consonant at *word-final position* always loses its inherent /ɔ/ in standard Dhaka Bangla — regardless of consonant type.

· Adding any suffix (case ending, aspect marker, article) shifts that consonant to non-final position and the inherent vowel rule (or matra rule) applies normally again.

· Formal, poetic, or Sanskrit-influenced speech may retain the final /ɔ/ — this is a register difference, not an error.

SUMMARY — ALL 13 BANGLA → IPA RULES

#	RULE NAME	WHAT IT DOES	IPA RESULT	WORKED EXAMPLE	KEY EXCEPTION / NOTE
1	**Independent vowels** স্বরবর্ণ	A standalone vowel letter maps directly to its IPA symbol	অ→/ɔ/ আ→/a/ ই→/i/ উ→/u/ এ→/e/ ও→/o/	আম → /am/	ই and ঈ both → /i/; উ and ঊ both → /u/
2	**Vowel signs (Matras)** কারচিহ্ন	A matra attached to a consonant replaces the inherent /ɔ/ with its own vowel	◌া→/a/ ◌ি→/i/ ◌ু→/u/ ◌ে→/e/ ◌ো→/o/ ◌ঁ→/◌̃/	কাজ → /kadʒ/	Matra always wins over inherent /ɔ/ — no competition
3	**4-way stop contrast** স্পর্শবর্ণ	Stops are distinguished by two independent features: voicing and aspiration	p pʰ b bʰ · t̪ t̪ʰ d̪ d̪ʰ · ʈ ʈʰ ɖ ɖʰ · tʃ tʃʰ dʒ dʒʰ · k kʰ g gʰ	ক খ গ ঘ → /k kʰ g gʰ/	5 places × 4 contrasts = 20 stop phonemes; ফ/ভ → /f/v/ in loanwords
4	**Nasals, fricatives, liquids**	These consonants map directly to IPA with no special rules	ন/ণ/ঞ→/n/ ম→/m/ ◌ং→/ŋ/ · শ/ষ→/ʃ/ স→/s/ হ→/h/ · র→/r/ ল→/l/	নাম → /nam/	ণ, ঞ, ষ collapse to dental equivalents in spoken Bangla
5	**Special signs** ড় ঢ় য় ◌ং ◌ঃ ◌ঁ ◌্	Modified consonants and diacritics with unique IPA values	ড়→/ɽ/ ঢ়→/ɽʰ/ য়→/j/ ◌ং→/ŋ/ ◌ঃ→/h/ ◌ঁ→◌̃ ◌্→Ø	নয় → /nɔj/	য → /dʒ/ initially, /j/ medially; ব → /b/ initially, /w/ medially
6	**Inherent vowel rule** অন্তর্নিহিত স্বর	Every bare consonant in non-final position automatically carries /ɔ/	C (bare, non-final) → C + /ɔ/	গরম → /gɔrɔm/	Overridden by matra (rule 2) or killed by hasanta (rule 9)
7	**Word-final vowel drop**	A bare consonant at word-final position loses its inherent /ɔ/	C (bare, final) → C only (no /ɔ/)	ঘর → /gʰɔr/ not /gʰɔrɔ/	Adding any suffix makes the consonant non-final → /ɔ/ reactivates
8	**Vowel harmony** স্বরসঙ্গতি	Inherent /ɔ/ raises to /o/ when the following syllable has /i, u, e, o/	/ɔ/ → /o/ / __ {i, u, e, o}	করি → /kori/ (not /kɔri/)	Blocked by /a/ or /ɔ/ in following syllable; applies only to inherent /ɔ/
9	**Hasanta (virama)** ◌্	Kills the inherent /ɔ/ of the consonant it sits under; leaves it bare	C + ◌্ → bare /C/ (no vowel)	শক্তি: ক + ◌্ → /k/	Without hasanta the same letters would produce extra syllables: শকতি = /ʃɔkɔt̪i/

10	**Conjunct consonants** যুক্তবর্ণ	Two consonants joined by hasanta fuse into a single cluster with no vowel between them	C_1 + ্ + C_2 → /C_1C_2/	**বিজ্ঞান**: জ + ্ + ঞ → /dʒn/	The vowel after the cluster belongs to C_2, not C_1
11	**Chandrabindu** ঁ	Nasalizes the vowel of the syllable it marks — adds no consonant of its own	V → Ṽ (tilde on vowel): /a/→/ã/ /e/→/ẽ/ /o/→/õ/	**চাঁদ** → /tʃãd̪/	Unlike anusvara, it adds no IPA segment — only a feature on the vowel
12	**Anusvara** ং	Adds a velar nasal /ŋ/ as a syllable coda — always closes a syllable, never opens one	ং → /ŋ/ (coda only)	**রং** → /rɔŋ/ · **বাংলা** → /baŋ.la/	Does not replace inherent /ɔ/ of preceding consonant — /ɔ/ still applies
13	**Visarga** ঃ	Adds /h/ after a vowel as a coda; merges into aspiration of a following aspirated stop	ঃ → /h/ post-vowel; ঃ + aspirated stop → stop only	**দুঃখ** → /d̪ukʰ/ · **অতঃপর** → /ɔt̪ɔhpɔr/	Before unaspirated consonant or word-finally → /h/ pronounced distinctly

Order of application for each consonant:

① Identify grapheme type (table lookup: rules 1–5)

② Has hasanta ্? → bare consonant, join with next (rules 9–10)

③ Has matra? → use matra's vowel, skip rule 6 (rule 2)

④ Bare & word-final? → no vowel at all (rule 7)

⑤ Bare & non-final? → assign inherent /ɔ/ (rule 6)

⑥ Check vowel harmony on each /ɔ/ assigned in step ⑤ (rule 8)

⑦ Apply diacritics: ঁ nasalizes vowel (rule 11) · ং adds /ŋ/ coda (rule 12) · ঃ adds /h/ coda (rule 13)

QUICK DECISION FLOWCHART

For each consonant in the word:

① Has a **matra** attached? → use matra's vowel (no inherent /ɔ/)

② Has **hasanta** ্? → no vowel at all (conjunct with next consonant)

③ Is it **word-final**? → no inherent vowel (dropped in standard Dhaka speech)

④ None of the above → append inherent /ɔ/

⑤ Check **vowel harmony**: if /ɔ/ is followed by /i, u, e, o/ → raise to /o/